Empório da Música — AI Customer Support Agent

Artefact — AI Engineer Technical Challenge (2026) | Technical Documentation & README

Public GitHub Repository:	https://github.com/vinialves2020/emporio-da-musica-ai-agent
Architecture:	Hybrid Function-Calling (ReAct) + Relational SQLite Engine + Section-Aware Policy RAG
LLM Providers Supported:	Groq API (Llama 3.3 70B), Local Ollama (Llama 3.2), OpenAI, Gemini + Built-in Offline Deterministic ReAct Engine
Interfaces & Test Suite:	Rich Interactive CLI + Streamlit WhatsApp Web UI | 14 Automated Pytest Unit & Integration Tests

1. Project Overview

Empório da Música (*'Sua música começa aqui.'*) is a specialized musical instrument store located in Campo Grande/MS, founded in 2008. To relieve the human customer support team from repetitive WhatsApp inquiries, this project implements a modular, production-grade **AI Customer Support Agent** in Python that:

• **Embodies the Store Persona (Section 7.1):** Informal yet professional, warm, and musical — communicating like a knowledgeable musician friend while maintaining strict factual grounding.

• **Dynamically Routes Between Operational Data & Store Policies:** Queries relational SQLite tables (*products, categories, promotions, orders, order_items, customers*) for live prices, stock, promotions, and order status, and queries the Store Manual (*data/políticas_da_loja.pdf*) via a Hybrid RAG retriever for store address, opening hours, payment/installment rules, shipping/freight, returns/exchanges, warranty, and LGPD privacy.

• **Enforces Strict Scope Guardrails (Sections 1 & 7.1):** Politely redirects requests for accessories (loose strings, picks, cables, cases, pedals, amplifiers) since the store sells exclusively musical instruments, and declines unrelated out-of-domain topics.

2. System Architecture & Repository Structure

The agent follows a clean separation of concerns across 4 layers: Data ETL & Relational Storage (**src/data/database.py**), Section-Aware Policy RAG (**src/retrieval/policy_rag.py**), Deterministic Domain Tool Schemas (**src/tools/store_tools.py**), and the ReAct Agent Orchestrator with SQLite Session Memory (**src/agent/engine.py** & **src/agent/memory.py**).

```
Customer (Rich CLI / Streamlit WhatsApp UI)
   |
   v
EmporioAgent Orchestrator (Hybrid ReAct Loop + SQLite SessionMemory)
   |
   +---> [Tool 1] search_products()               -> Relational SQLite + RapidFuzz + Specs Filter
   +---> [Tool 2] get_promotions()                -> Active vs. Expired Campaigns + Section 6.2 Rules
   +---> [Tool 3] lookup_order_or_customer()      -> Order Tracking + 7d/30d/90d Eligibility Math
   +---> [Tool 4] calculate_payment_and_shipping()-> PIX 5% + 3x/6x/12x Tiers + Motoboy/Correios
   +---> [Tool 5] search_store_policies()         -> Hybrid TF-IDF (Word + Char N-Grams) Policy RAG
```

3. Quickstart & Execution Instructions

Requires **Python 3.10+**. Works out-of-the-box in both cloud/local LLM mode (Groq, Ollama, OpenAI, Gemini) and zero-configuration offline mode:

```
# 1. Clone the repository and install dependencies
git clone https://github.com/vinialves2020/emporio-da-musica-ai-agent.git
cd emporio-da-musica-ai-agent
pip install -r requirements.txt

# 2. Optional: configure .env (Groq / Ollama / OpenAI / Gemini)
cp .env.example .env

# 3. Run the 5 automated showcase scenarios OR interactive CLI / Streamlit UI
python main.py --demo
python main.py
streamlit run src/ui/streamlit_app.py

# 4. Run the automated test suite (14 unit & integration tests)
pytest
```

4. Justification of Technical Decisions

Decision Dimension	Chosen Approach & Engineering Justification
Agent Architecture *(Hybrid ReAct + SQL + RAG)*	Pure vector RAG fails on numeric filters (*'violões até R$ 1000'*), stock checks, and multi-table joins. Conversely, raw LLM Text-to-SQL risks syntax errors and missed business rules (e.g., non-cumulative PIX discount). Exposing **5 deterministic domain tools** over SQLite + Hybrid RAG guarantees 100% factual and arithmetic precision.
Model & Provider *(Groq / Ollama + Offline Engine)*	Configured for **Groq (llama-3.3-70b-versatile)** and **Local Ollama (llama3.2)** via the standard OpenAI SDK (plus OpenAI/Gemini support). Also includes a built-in **Deterministic Offline ReAct Engine** that executes the exact same tools when no API key is configured, ensuring zero-friction evaluation.
Policy RAG Strategy *(Section-Aware Dual TF-IDF)*	*políticas_da_loja.pdf* contains structured tables (payment tiers, shipping carriers) that break under naive fixed-length chunking. We extract the PDF via *pypdf* into 10 semantic sections and index them with a **Word (1-2 ngram) + Character (3-5 ngram) TF-IDF + Bilingual Tag Booster** (<5ms latency, zero external embedding cost).
Session Persistence *(SQLite SessionMemory)*	Backed by SQLite (*sessions.db*) in *src/agent/memory.py* so customers can naturally reference previous turns (e.g., asking if they can return an item in Turn 1, and providing Order #8 in Turn 2).
User Interfaces *(Rich CLI + Streamlit UI)*	Provides both a fast terminal CLI (with live tool-trace inspection tables and *--demo* mode) and a WhatsApp-themed Streamlit web application with quick-test buttons and an expandable JSON tool inspector.

5. Data Cleaning & Exploratory Audit (Documented Assumptions)

During our exploratory audit of the 6 CSV files and the 8-page policy manual, we identified and programmatically resolved 7 real-world data anomalies:

• **1. Active Product with Zero Stock (product_id=96):** *Giannini GF-3D Dreadnought Sunburst* (R$ 799.90) has *status='active'* but *stock_quantity=0*. When filtering for available stock (*only_in_stock=True*), the query enforces both *status=='active'* and *stock_quantity > 0* (returning the 12 truly available guitars under R$ 1,000), and adds a transparency note citing Section 7.3.

• **2. Discontinued & Coming Soon Statuses (IDs #113, #136, #130):** *Shelby SN-7C 7 Cordas* (#113) is *discontinued* (stock=0); *Ibanez Bass 2X* (#136) has *status='discontinued'* despite *stock_quantity=13*; *Kala KA-ABP-CTG Barítono Cedar* (#130) has *status='coming_soon'* (and appears in orders #3 and #14). The agent explicitly reports these statuses and suggests active alternatives.

• **3. Active vs. Expired Promotions & Non-Cumulative PIX Rule (promotions.csv):** Only 4 of the 25 promotions have *is_active == 1* (#12 Kalani KAL-700T 10% OFF, #14 Ohana CK-20 20% OFF, #19 Crafter HT-100 18% OFF, #24

Taylor 110e 8% OFF). The remaining 21 campaigns (including Black Friday, Semana do Músico, and #4 on Takamine GD20) have *is_active == 0*. Per Section 6.2, the 5% PIX discount is never stacked onto active promotional prices.

• **4. Empty Categories in CSV (Categories 6, 7, and 8):** *categories.csv* lists 9 categories, but *products.csv* has 0 items for *6: Sopro (Madeiras)*, *7: Sopro (Metais)*, and *8: Cordas Orquestrais*. The database layer detects empty category queries and informs the customer transparently.

• **5. Synthetic Brand/Spec Mismatches in Products #135–#144:** Rows #135–#144 have different brands in their title vs. description (e.g., #135 title says *Music Man Bass 1X* while description says *Fender Jazz Bass*). Both fields are indexed for search and annotated with a catalog note.

• **6. Contact & Email Normalization in Manual:** Page 2 lists landline/general WhatsApp *(67) 3341-4444* and an accented email, while Page 1 lists *contato@emporiodamusica.com.br* and Page 6 Section 7 lists remote WhatsApp *(67) 3321-4500*. The agent presents both official phone numbers and the valid ASCII email.

• **7. Deterministic Reference Date (REFERENCE_DATE=2026-03-23):** Since dataset orders span Oct 2025 – Mar 2026 (latest order #16 on 2026-03-22), setting the default reference date to *2026-03-23* ensures deterministic evaluation of 7-day regret and 30-day defect windows.

6. Summary of Included Agent Interaction Examples (examples/)

Example File	Customer Prompt / Scenario	Tools Invoked & Policy/Data Rules Demonstrated
01_catalog_violoes_under_1000.md	*'Quais opções de violões disponíveis custando até R$1000?'* + GF-3D follow-up	Calls **search_products**. Lists all 12 in-stock guitars <= R$1000 with PIX 5% & installments, while noting #96 (0 stock) and #113 (discontinued) + alternatives.
02_store_info_hours_and_shipping.md	*'Qual o endereço da loja e horário?'* + Frete Dourados & LGPD	Calls **search_store_policies**. Returns Rua 14 de Maio 3200, hours (Mon-Fri 9-18h, Sat 9-13h), free shipping > R$500 in CG, PAC/SEDEX/Jadlog, and LGPD.
03_price_inquiry_and_promotions.md	*'Quanto custa o Takamine GD20?'* + Active promotions & PIX rule	Calls **search_products**, **calculate_payment_and_shipping**, and **get_promotions**. Shows R$ 2,199 (R$ 2,089.05 PIX, 12x R$ 183.25, combined payment) + 4 active promos.
04_order_tracking_and_return_policy.md	*'Me arrependi da minha compra, posso devolver meu pedido?'* (Orders #8 & #7)	Calls **lookup_order_or_customer** + **search_store_policies**. Evaluates Order #8 (*shipped*, eligible on arrival) vs Order #7 (*delivered* 34d ago, 90d CDC active).
05_out_of_scope_and_accessories_guardrails.md	Accessory inquiry (strings, picks, amp) + Unrelated recipe/weather prompt	Enforces Sections 1 & 7.1 (instruments-only scope, redirects accessories to partner shops) and politely declines out-of-domain chitchat.

7. Known Limitations & Future Roadmap

• **1. Read-Only Order & Complaint State:** The agent inspects live orders and explains the 24-business-hour SLA for complaints (Section 7.3), but does not mutate *orders.csv*. *Next step:* Add write-enabled transactional tools backed by PostgreSQL and CRM/ERP webhooks.

• **2. Real-Time Carrier Freight API:** For cities outside Campo Grande, the agent reports the official PAC (5–12bd), SEDEX (2–5bd), and Jadlog (3–8bd) timeframes and insurance rules. *Next step:* Integrate Correios/Melhor Envio APIs for exact CEP-based BRL freight quotes.

• **3. Customer 2FA Verification:** Currently, order lookup by ID or name returns order status directly (with disambiguation warnings for duplicate names like *Diego Castro* or *Bruno Martins*). *Next step:* Require matching phone/CPF digits before revealing tracking data per LGPD.

• **4. WhatsApp Business Cloud API Webhook:** *Next step:* Expose *EmporioAgent* via a FastAPI webhook connected to Meta's WhatsApp Cloud API with automatic *America/Campo_Grande* (UTC-4) business-hours detection.

8. Use of AI Coding Assistants

Tool Used: Google Antigravity (AI Coding Assistant).
Workflow: Used to (1) cross-audit all 6 CSV files against the 8 pages of *políticas_da_loja.pdf* to uncover edge cases (#96 zero stock, #135–#144 brand mismatches, 4 active vs. 21 expired promotions); (2) design the 5 deterministic tool schemas in *StoreToolkit* so financial and date rules are computed without hallucination; and (3) build and run the 14 automated *pytest* tests verifying all scenarios.